

Purpose

The DIAL Adaptive Screening Engine is an AI-assisted decision support system that identifies learners who may be at risk of specific learning differences and recommends the **single most appropriate assessment pathway**.

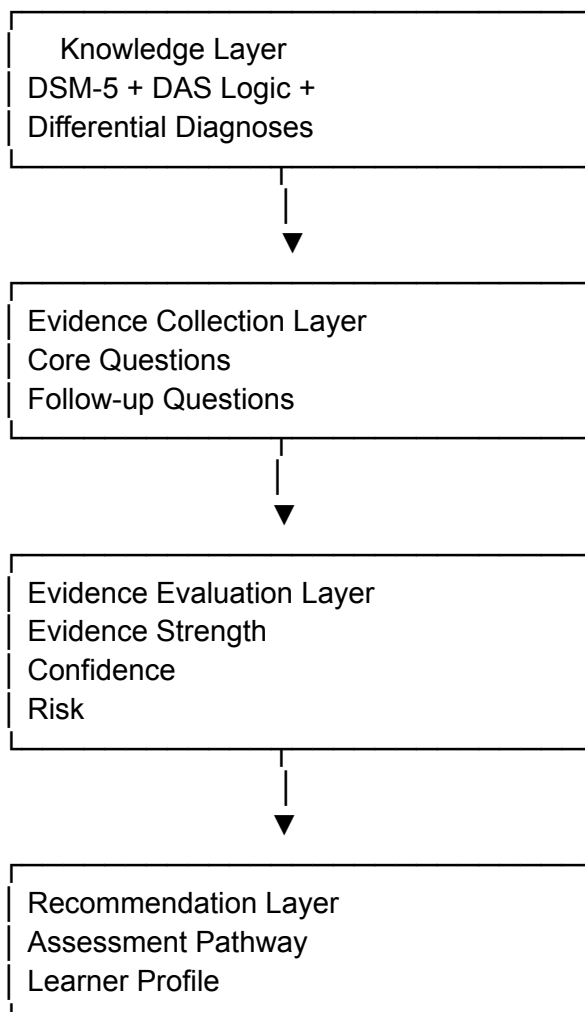
The screening engine **does not diagnose** learning disorders. Instead, it gathers behavioural evidence through adaptive questioning, estimates the likelihood of different learning profiles, and recommends an appropriate next step.

The output serves three purposes:

1. Recommend an assessment pathway.
2. Create an initial learner profile for DIAL.
3. Reduce unnecessary referrals by identifying the primary presenting concern.

Our Proposed Architecture

The four layers are as follows:



Initially, we thought:

Question → Score → Recommendation

Now, we're thinking of something closer to:

**Question → Gather evidence → Compare against clinical knowledge →
Decide what evidence is still missing → Ask another question**

Layer 1: Knowledge Layer

Source	Purpose
DSM-5-TR Diagnostic Criteria	Defines the hallmark characteristics of each disorder.
DSM-5-TR Differential Diagnosis	Helps distinguish between disorders with overlapping behaviours.
DSM-5-TR Descriptive Features	Gives AI richer understanding to ask better follow-up questions.
DAS Referral Logic	Determines when an assessment should be recommended.
Psychologist Decision Rules	Human-approved operational rules for scoring and referral.
Question Bank	Behaviour-based questions derived from the above sources.

DSM isn't used to score. DSM is used to teach the interviewer.

Layer 2: Evidence Collection

Instead of thinking: “Ask Question 1”

Think: “Collect Evidence Type A.”

Evidence Type	Example Behaviour
Reading Accuracy	Skips words
Decoding	Cannot sound out unfamiliar words
Reading Fluency	Reads slowly
Attention Regulation	Easily distracted
Executive Function	Loses belongings
Mathematical Processing	Cannot understand place value

Layer 3: Evidence Evaluation

This is where I think DIAL becomes unique.

Instead of producing “Dyslexia Score = 42”, the engine should produce something like:

Evidence Category	Strength	Confidence
Decoding	Strong	High
Reading Fluency	Moderate	High
Spelling	Strong	Medium
Working Memory	Weak	High

Layer 4: Recommendation

Only now does the system decide

1. Primary concern
2. Secondary concern
3. Risk level
4. Recommended assessment
5. Reasoning

Now let's think about AI.

We actually think we need **three different AI responsibilities**.

AI Role 1:

Conversation Manager

Its only job is

- ask questions
- clarify answers
- simplify wording

Nothing else.

AI Role 2:

Evidence Investigator

Suppose parent says

"My child hates reading."

AI knows that reading avoidance alone is insufficient.

So it asks

Does your child avoid reading because reading is difficult, because they lose attention quickly, or because they simply dislike reading?

That's what differential diagnosis is.

AI Role 3

Report Writer

Once scoring is complete

AI writes

- learner profile
- explanation
- recommendation

And now, let's look at TWO scoring systems:

Score 1

Evidence Strength

This measures how much evidence exists.

Score 2

Confidence

This measures how reliable the evidence is.

Example A:

Parent answers "I don't know."

Confidence drops.

Example B:

Parent gives three contradictory answers.

Confidence drops.

Example C:

Teacher and parent both report the same behaviour.

Confidence increases.

Differential diagnosis

Suppose the child avoids reading.

Possible explanations

- Dyslexia
- ADHD
- Anxiety
- English not first language

The AI should ask questions specifically to eliminate alternatives because that's how psychologists work.

Suggested workflow

Start



Consent



Basic Profile



Core Evidence Collection



Initial Evidence Analysis



Possible Domains Identified



AI Consults Knowledge Base



AI Performs Differential Investigation



Evidence Updated



Confidence Calculated



Risk Determined



Recommendation



Learner Profile



Human Review

The AI doesn't simply ask more questions.

It asks questions because there is **uncertainty**.

This is therefore technically an adaptive reasoning screener engine.

Because it doesn't just total points.

It mimics the early reasoning process of a psychologist:

- collect evidence,
- test hypotheses,
- rule out alternatives,
- identify the most likely primary concern,
- explain why,
- recommend the most appropriate next step.

Of course, for users and project documentation, "Adaptive Screening Engine" remains the right public-facing name. But as a design philosophy, thinking of it as an adaptive clinical reasoning engine will lead to a much stronger architecture and clearer implementation

Design Blueprint

Order	Task	Why
1	Define the knowledge base structure	So we know what the AI can refer to.
2	Define the evidence categories	So every question maps to a skill/behaviour area.
3	Create the evidence strength framework	Core / Supporting / Contextual / Not Relevant.
4	Create the adaptive flow	So the screener knows when to branch.
5	Create the question bank	Core questions + Dyslexia + ADHD + Dyscalculia branches.
6	Create the scoring and recommendation rules	So output is consistent and explainable.
7	Create the JSON output format	So ProbStatement1 can connect to the DIAL Profiling Engine.

Step 1: Knowledge Base Specification

Knowledge Base Component	What It Contains	How DIAL Uses It
DSM-5-TR Diagnostic Criteria	Formal criteria for Dyslexia/Specific Learning Disorder, ADHD, and Dyscalculia/SLD with mathematics impairment.	Used to derive behavioural indicators, not to diagnose.
DSM-5-TR Descriptive Features	How each condition commonly presents in real life.	Helps AI understand user responses and ask better follow-up questions.
DSM-5-TR Differential Diagnosis Considerations	Overlapping conditions and alternative explanations.	Helps AI avoid premature conclusions and ask critical clarification questions.
DAS Referral Logic	DAS-specific rules on when to recommend assessment, KIV secondary concerns, or suggest monitoring.	Converts screening evidence into an appropriate assessment pathway.
Psychologist-Approved Decision Rules	Rules validated by DAS psychologists, including red flags and human review triggers.	Keeps AI bounded, safe, and auditable.
Behavioural Question Bank	Parent/student/teacher-friendly questions based on observable behaviours.	Used for core questions, domain-specific branches, and secondary probes.
Evidence Mapping Table	Maps every question to Dyslexia, ADHD, and Dyscalculia using Evidence Strength.	Enables deterministic scoring and explainable recommendations.
Output Template	Standard wording for risk level, primary concern, secondary concern, next step, and learner profile.	Ensures consistent, non-diagnostic language.

The knowledge base should guide AI questioning, but final scoring must remain deterministic and based on psychologist-approved rules. The AI may gather and summarise evidence, but it must not independently diagnose, override evidence weights, or recommend multiple assessment pathways without human review.

This table is not the question bank. Instead, think of it as the ontology (or vocabulary) of the screening engine. Every question we eventually write will map to one or more of these evidence categories.

For example: "Has difficulty sounding out unfamiliar words." doesn't map directly to Dyslexia.

It maps to the evidence category: Phonological Processing, which is the evidence category then contributes evidence towards Dyslexia. This makes the whole system modular and future-proof.

Behavioural Evidence Categories Framework

Purpose

The DIAL Adaptive Screening Engine collects behavioural evidence rather than attempting to diagnose learning differences directly.

Each question is designed to gather behavioural evidence for one or more behavioural domains (Behaviour Evidence Categories). These categories represent observable learning behaviours commonly associated with different learning differences and are derived from DSM-5-TR diagnostic criteria, descriptive features, differential diagnosis considerations, and psychologist-approved decision rules.

Rather than asking, "Does the learner have dyslexia?", the system asks, "What evidence do we have about this learner's reading, attention, executive functioning, and mathematical learning?"

Evidence Categories

Category ID	Behavioural Evidence Category	Description	Commonly Associated With
BEC01	Phonological Processing	Ability to recognise, manipulate, and remember speech sounds.	Dyslexia
BEC02	Word Reading Accuracy	Accuracy when reading familiar and unfamiliar words.	Dyslexia
BEC03	Reading Fluency	Speed, effort, and smoothness of reading.	Dyslexia
BEC04	Reading Comprehension	Understanding and interpreting written text.	Dyslexia (secondary), ADHD
BEC05	Spelling & Written Language	Ability to spell consistently and express ideas in writing.	Dyslexia

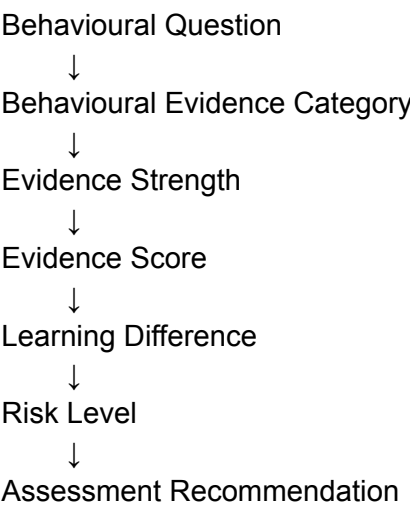
BEC06	Attention Regulation	Ability to sustain attention and resist distractions.	ADHD
BEC07	Impulsivity	Acting before thinking, interrupting others, difficulty waiting.	ADHD
BEC08	Hyperactivity	Excessive movement, fidgeting, restlessness.	ADHD
BEC09	Executive Functioning	Planning, organisation, working memory, task completion.	ADHD, Dyslexia
BEC10	Working Memory	Holding and manipulating information while completing tasks.	Dyslexia, ADHD, Dyscalculia
BEC11	Numerical Processing	Understanding numbers, quantity, and number relationships.	Dyscalculia
BEC12	Arithmetic Skills	Learning and recalling basic mathematical facts and procedures.	Dyscalculia
BEC13	Mathematical Reasoning	Applying mathematical concepts to solve problems.	Dyscalculia
BEC14	Learning Behaviours	Persistence, task avoidance, frustration, motivation.	All domains (Contextual)
BEC15	Developmental & Family History	Previous concerns, family history, developmental milestones.	All domains (Contextual)

Why Categories Instead of Direct Disorder Mapping? This design provides several advantages:

Traditional Screener	DIAL Approach
Questions point directly to a disorder.	Questions first gather behavioural evidence.
Difficult to reuse questions.	Questions can support multiple learning differences.
Less transparent reasoning.	Every recommendation can be traced through evidence categories.
Harder to expand to new disorders.	New learning differences can reuse existing evidence categories.

Evidence Flow

The following diagram illustrates how information flows through the screening engine.



Example Mapping

Behavioural Question	Evidence Category	Evidence Strength	Associated Learning Difference
Has difficulty sounding out unfamiliar words.	BEC01 Phonological Processing	Core	Dyslexia
Frequently skips words while reading.	BEC03 Reading Fluency	Supporting	Dyslexia
Frequently loses homework or school materials.	BEC09 Executive Functioning	Core	ADHD
Constantly fidgets during lessons.	BEC08 Hyperactivity	Core	ADHD
Struggles to understand place value.	BEC11 Numerical Processing	Core	Dyscalculia
Homework often causes frustration.	BEC14 Learning Behaviours	Contextual	All domains

The Adaptive Reasoning Cycle

Instead of this:

Question



Score



Next Question

Collect Evidence



Evaluate Evidence



Identify Uncertainty



Generate Hypotheses



Ask Most Valuable Question



Update Evidence



Repeat until confidence threshold reached

Notice the difference.

The AI isn't asking random follow-up questions.

It is asking the next best question.

Screening Engine's Reasoning Loop		
Stage	What Happens	Responsible Component
1. Collect	User answers behavioural question.	User
2. Interpret	Response mapped to Behavioural Evidence Categories (BECs).	Rule Engine
3. Evaluate	Evidence strengths and confidence updated.	Rule Engine
4. Hypothesise	Possible learning profiles ranked (e.g., Dyslexia > ADHD > Dyscalculia).	Decision Engine
5. Identify Gaps	Determine what evidence is still missing or conflicting.	Decision Engine
6. Ask Next Best Question	AI selects the most informative follow-up question.	LLM
7. Repeat	Continue until sufficient evidence has been collected.	Entire System

Decision Rules

We also need to define when to stop asking questions. This prevents the AI from endlessly asking questions.

Condition	Action
Evidence confidence $\geq 90\%$	Stop screening
Maximum questions reached	Stop screening
One domain clearly dominates	Stop screening
Evidence remains ambiguous	Recommend psychologist review
User exits early	Generate partial report with lower confidence

Question Prioritisation

Every unanswered question has a value. The system should estimate: "If I ask this question, how much will it reduce uncertainty?" Then ask the highest-value question. For example:

Candidate Question	Expected Value
Reading fluency	High
Handwriting	Medium
Family history	Low

The AI chooses Reading Fluency. This is a concept borrowed from decision science called information gain.

Question States

Every question in the bank should have metadata. This turns the question bank into an adaptive decision tree instead of a fixed list.

Field	Example
Question ID	Q023
Behavioural Evidence Category	BEC03
Difficulty	Parent-friendly
Target Domain	Dyslexia
Evidence Strength	Core
Trigger Conditions	Only ask if Reading evidence is Moderate or High
Follow-up Questions	Q024, Q025
Stop Conditions	Skip if confidence already High

Question Bank

Each question will have a structure like this:

Field	Value
Question ID	Q001
Behavioural Question	My child avoids reading aloud.
Behavioural Evidence Category	BEC03 Reading Fluency
Primary Association	Dyslexia
Secondary Association	None
Evidence Strength	Core
Response Scale	Never → Almost Always
Trigger	Core Screening
Follow-up Trigger	If response ≥ Often
Clarification Prompt	Can you describe what happens when your child reads aloud?

Decision Engine: Core Idea

Part	Function
Evidence Tracker	Keeps running scores for Dyslexia, ADHD, and Dyscalculia.
Confidence Tracker	Checks whether the evidence is reliable enough.
Uncertainty Detector	Identifies unclear, contradictory, or overlapping evidence.
Adaptive Router	Decides which branch or follow-up question should come next.
Primary Concern Selector	Chooses the strongest presenting issue.
Secondary Concern Handler	Flags secondary concerns as KIV instead of recommending multiple assessments.
Human Review Trigger	Sends complex cases for psychologist/admin review.

Simple Decision Flow

Stage	Decision Question	Possible Action
1. After core questions	Is one domain clearly strongest?	Route to that domain branch.
2. If mixed evidence	Are two domains close?	Ask differential follow-up questions.
3. After branch questions	Is primary evidence strong enough?	Assign Low / Moderate / High risk.
4. Secondary check	Is another concern meaningful but not dominant?	Mark as KIV.
5. Confidence check	Are answers consistent and complete?	Proceed, ask clarification, or human review.
6. Final decision	Is one assessment pathway most appropriate?	Recommend one primary assessment.

The Key Rule

Scenario	Engine Decision
Dyslexia much higher than ADHD/Dyscalculia	Recommend Dyslexia assessment.
ADHD much higher than Dyslexia/Dyscalculia	Recommend ADHD assessment.
Dyscalculia much higher than others	Recommend Dyscalculia assessment.
Dyslexia high + ADHD moderate	Recommend Dyslexia assessment; ADHD as KIV.
ADHD high + Dyslexia moderate	Recommend ADHD assessment; Dyslexia as KIV.
Two domains equally high	Trigger human review or ask differential questions.
All domains low	No assessment recommended yet; monitor/support.
All domains moderate	Recommend consultation/human review, not automatic assessment.

Hybrid Engine:

Layer	Method
Scoring	Deterministic evidence-strength scoring.
Routing	Rule-based adaptive branching.
Clarification	AI asks bounded follow-up questions.
Recommendation	Rule engine decides; AI only explains.

Decision Engine Logic

Input: User responses

↓

Map each answer to Behavioural Evidence Categories

↓

Convert Evidence Strength into score

↓

Update domain evidence:

- Dyslexia
- ADHD
- Dyscalculia

↓

Check confidence:

- complete?
- consistent?
- enough core indicators?
- contradictions?

↓

Decide next action:

- A. ask more questions
- B. branch to domain-specific screener
- C. run secondary probe
- D. trigger human review
- E. generate recommendation

Decision Engine Output

Output Field	Example
Primary Concern	Dyslexia indicators strongest
Secondary Concern	Attention regulation concerns to KIV
Risk Level	High
Confidence Level	Medium-high
Recommended Pathway	Dyslexia assessment
Reasoning	Strong evidence in decoding, reading fluency, spelling consistency
Human Review Needed?	No / Yes

Evidence Strength & Decision Framework

Purpose

The DIAL Adaptive Screening Engine evaluates learners by accumulating **behavioural evidence**, rather than simply summing questionnaire scores.

Each behavioural response contributes evidence towards one or more learning difference domains. The Decision Engine evaluates:

- the **strength** of the evidence,
- the **consistency** of the evidence,
- the **confidence** of the evidence,
- and the **presence of hallmark indicators**

before recommending an assessment pathway.

1. Evidence Strength Levels

Each behavioural question is assigned an Evidence Strength for every learning difference.

Evidence Strength	Internal Weight	Definition	Example
Core Indicator	4	Highly characteristic behaviour that strongly supports a particular learning difference.	Difficulty decoding unfamiliar words
Supporting Indicator	2	Frequently associated behaviour that strengthens existing evidence but is not unique to one disorder.	Difficulty following written instructions
Contextual Indicator	1	General behavioural information that provides context but is not diagnostic on its own.	Homework often causes frustration
Not Relevant	0	Behaviour does not meaningfully contribute towards the learning difference.	Reading avoidance for Dyscalculia

2. Behavioural Response Scale

All behavioural questions use the same response scale.

Response	Score
Never	0
Rarely	1
Sometimes	2
Often	3
Almost Always	4

Evidence contribution is calculated internally
as:

$$\text{Evidence Contribution} = \text{Response Score} \times \text{Evidence Weight}$$

3. Evidence Rules

Instead of relying purely on accumulated scores, DIAL applies a series of evidence-based decision rules.

Rule 1: Hallmark Behaviours Must Be Present

A learner **cannot** be classified as **High Risk** for a learning difference unless there is sufficient evidence from **Core Indicators**.

Example:

A learner reports:

- frustration
- slow homework
- poor concentration

These alone should **not** trigger a High Risk Dyslexia recommendation because they are not hallmark reading behaviours.

Rule 2: Supporting Evidence Increases Confidence

Supporting Indicators strengthen confidence in an existing hypothesis but cannot independently determine the primary concern.

Example:

Difficulty remembering spelling patterns

becomes more meaningful when combined with

- poor decoding
- inaccurate word reading
- slow reading fluency

Rule 3: Context Provides Explanation

Contextual Indicators help explain learning behaviours but should never determine risk levels independently.

Examples include:

- avoids homework
- low confidence
- frustration
- family history

These help interpret the learner profile rather than identify the learning difference itself.

Rule 4: Evidence Must Be Consistent

The engine checks for consistency across responses.

Example

Parent reports "Reads fluently"

Later reports "Frequently unable to recognise simple words"

The Decision Engine detects inconsistency. Instead of producing a recommendation immediately. AI asks clarification questions.

Rule 5: Differential Evidence

When evidence supports multiple possible explanations, the engine attempts to reduce uncertainty.

Example

Reading avoidance may result from

- Dyslexia
- ADHD
- anxiety
- English language proficiency
- lack of motivation

Rather than assuming Dyslexia, AI asks questions designed to distinguish these possibilities.

This mirrors the differential diagnosis process used by psychologists.

4. Risk Determination Framework

Risk is determined using **patterns of evidence**, not numerical totals alone. Notice that **High Risk** is defined by the quality of evidence, not simply the quantity.

Risk Level	Characteristics
Low Risk	Few indicators, weak evidence, or isolated contextual behaviours.
Moderate Risk	Several Supporting Indicators and/or limited Core Indicators. Additional monitoring or discussion may be appropriate.
High Risk	Multiple Core Indicators supported by consistent Supporting Indicators and minimal contradictory evidence.

5. Confidence Framework

Risk and confidence are evaluated independently.

Confidence Level	Meaning
High	Responses are complete, internally consistent, and supported by multiple behavioural indicators.
Medium	Some uncertainty remains but sufficient evidence exists for a recommendation.
Low	Evidence is incomplete, inconsistent, or contradictory. Additional clarification or human review is recommended.

This distinction allows DIAL to report, for example:

- **High Risk + High Confidence** → Recommend assessment.
- **High Risk + Low Confidence** → Recommend psychologist review before referral.
- **Moderate Risk + High Confidence** → Consider monitoring or targeted support.

6. Primary Concern Selection

The Decision Engine identifies the learning difference with the strongest overall evidence. Only **one assessment pathway** is recommended automatically.

This aligns with DAS operational workflows and reduces unnecessary combination assessment referrals.

Scenario	Outcome
One domain clearly stronger	Recommend that assessment pathway.
Secondary domain has meaningful evidence	Record as Secondary Concern (KIV) .
Two domains equally strong	Trigger differential investigation or psychologist review.
Evidence remains unclear	Generate a low-confidence recommendation and recommend human review.

7. Human Review Triggers

The Decision Engine automatically requests psychologist or administrator review when one or more of the following conditions occur:

Trigger	Reason
Two or more domains have comparable evidence	Primary concern cannot be confidently determined.
Confidence level is Low	Evidence quality is insufficient for reliable recommendation.
Responses contain significant contradictions	Clarification did not resolve inconsistencies.
Red-flag behaviours are identified	Immediate professional review required.
User exits before completion	Partial evidence only.

8. Decision Philosophy

The DIAL Adaptive Screening Engine follows five guiding principles:

Principle	Description
Evidence before scores	Recommendations are driven by behavioural evidence rather than raw numerical totals.
Quality before quantity	Hallmark indicators carry greater significance than numerous general behaviours.
Explainability	Every recommendation can be traced back to the evidence that supports it.
Human oversight	AI supports evidence collection; psychologists retain responsibility for diagnostic decisions.
Adaptive investigation	AI asks additional questions only when they meaningfully reduce uncertainty.

Question Metadata Schema

Defining the structure each question must follow:

Field	Purpose
Question ID	Unique identifier, e.g. CQ01, DYX01, ADHD01, DCAL01.
Question Text	Parent/student/teacher-friendly behavioural question.
Audience	Parent, teacher, student, or all.
Behavioural Evidence Category	Maps to BEC codes, e.g. Reading Fluency, Attention Regulation.
Target Domain	Dyslexia, ADHD, Dyscalculia, or cross-domain.
Evidence Strength	Core, Supporting, Contextual, or Not Relevant.
Response Scale	Never, Rarely, Sometimes, Often, Almost Always.
Trigger Condition	When this question should appear.
Follow-up Trigger	When AI should ask a clarification question.
Clarification Prompt	Approved AI follow-up wording.
Red Flag?	Whether this answer should trigger human review.
Output Use	Whether the answer feeds assessment recommendation, learner profile, or both.

What evidence do we want?

Instead of writing questions randomly, we'll begin with the evidence we want first.

Behavioural Evidence Category	Purpose
Reading	Is literacy the main concern?
Writing	Are written language skills affected?
Mathematics	Is numeracy the main concern?
Attention	Does the learner sustain attention?
Executive Function	Organisation, planning, working memory
Learning Behaviour	Frustration, persistence, avoidance
Developmental Context	Family history, previous intervention

Core Question Types

Q	Behavioural Evidence Category	Purpose
CQ01	Reading	Reading difficulty
CQ02	Reading	Reading avoidance
CQ03	Writing	Spelling/writing
CQ04	Mathematics	Maths learning
CQ05	Attention	Sustained attention
CQ06	Executive Function	Organisation
CQ07	Working Memory	Multi-step instructions
CQ08	Learning Behaviour	Homework frustration
CQ09	Learning Behaviour	Time needed to complete work
CQ10	Reading	Reading confidence

CQ11	Mathematics	Number confidence
CQ12	Attention	Distractibility
CQ13	Executive Function	Forgetfulness
CQ14	General Learning	Compared with peers
CQ15	Overall Concern	User's biggest concern

Example Core Question

Field	Value
Question ID	CQ01
Behavioural Evidence Category	BEC02 Word Reading Accuracy
Purpose	Detect broad reading difficulty
Question	The learner finds reading more difficult than classmates of a similar age.
Audience	Parent / Teacher
Student Version	I find reading harder than most people in my class.
Response Scale	Never → Almost Always
Evidence Mapping	Dyslexia (Supporting), ADHD (Contextual), Dyscalculia (Not Relevant)

CQ02: The learner avoids reading whenever possible.
CQ03: The learner has difficulty spelling familiar words correctly.
CQ04: The learner finds mathematics more difficult than classmates of a similar age.
CQ05: The learner has difficulty staying focused on schoolwork without frequent reminders.
CQ06: The learner frequently loses school materials or forgets what they need to bring.
Cq07: The learner has difficulty remembering several instructions given one after another.
CQ08: Homework often becomes frustrating or emotionally upsetting.
CQ09: The learner takes much longer than peers to complete schoolwork.
CQ10: The learner lacks confidence when asked to read aloud.
CQ11: The learner lacks confidence when working with numbers or mathematics.
CQ12: The learner is easily distracted by surrounding activities or noises.
CQ13: The learner frequently forgets homework, instructions, or appointments.
CQ14: The learner seems to require significantly more support than classmates to keep up with schoolwork.
CQ15: Which area concerns you the most? Options • Reading • Writing • Mathematics • Attention • Organisation • Behaviour • I'm not sure This helps the AI understand the parent's mental model.
Core Questions ↓ Top 2 hypotheses ↓ AI consults Knowledge Base ↓ AI asks 3–5 differential questions ↓ Select primary branch

Soofrina's take:

The screening engine is already collecting the raw ingredients.

For example

Instead of storing

"Dyslexia Risk": 84

Store

```
{  
  "Reading Accuracy": "Low",  
  "Reading Fluency": "Low",  
  "Phonological Processing": "Needs Investigation",  
  "Working Memory": "Moderate",  
  "Attention Regulation": "Typical",  
  "Learning Behaviour": "Avoids Reading"  
}
```

Now imagine the lesson planner later...

It doesn't care about dyslexia.

It cares that

- decoding is weak
- phonological awareness is weak
- attention is fine
- confidence is low

That becomes the **initial learner model**.

Dyslexia Investigation Framework

Investigation Area	BEC	Clinical Purpose	Evidence Strength
1. Phonological Processing	BEC01	Check whether the learner struggles with sounds in words.	Core
2. Word Reading Accuracy	BEC02	Check whether the learner reads printed words accurately.	Core
3. Decoding Unfamiliar Words	BEC01/BEC02	Check whether the learner can sound out new words.	Core
4. Reading Fluency	BEC03	Check whether reading is slow, effortful, or hesitant.	Core
5. Spelling Consistency	BEC05	Check whether spelling difficulties are persistent and patterned.	Core
6. Written Expression	BEC05	Check whether writing output is affected by spelling, sequencing, or sentence construction.	Supporting
7. Reading Comprehension Impact	BEC04	Check whether comprehension is affected because decoding is effortful.	Supporting
8. Reading Avoidance & Emotional Impact	BEC14	Check whether the learner avoids literacy tasks due to difficulty or frustration.	Contextual / Supporting
9. Developmental & Family History	BEC15	Check whether difficulties are persistent, long-standing, or family-linked.	Contextual

Investigation Details

Area	Behavioural Questions	Differential Questions AI May Ask
Phonological Processing	"The learner has difficulty hearing or identifying sounds in words." / "The learner confuses similar-sounding words."	"Does this happen even with familiar spoken words?" / "Are there any hearing or speech-language concerns?"
Word Reading Accuracy	"The learner often misreads familiar words." / "The learner guesses words instead of reading them carefully."	"Does this happen even when the learner is focused?" / "Are the errors visual, sound-based, or random?"
Decoding Unfamiliar Words	"The learner struggles to sound out new or unfamiliar words." / "The learner relies on guessing when reading unknown words."	"Can the learner break the word into sounds?" / "Does the learner use phonics knowledge independently?"
Reading Fluency	"The learner reads slowly compared with peers." / "Reading aloud sounds hesitant or effortful."	"Is the reading slow because of word recognition, loss of attention, anxiety, or lack of practice?"
Spelling Consistency	"The learner spells the same word differently within the same piece of writing." / "The learner forgets spelling patterns even after practice."	"Are the spelling errors phonetic, visual, or careless?" / "Does spelling improve when the learner copies from a model?"
Written Expression	"The learner has good ideas but struggles to write them down." / "Written work is shorter or simpler than oral responses."	"Is the difficulty mainly spelling, sentence construction, handwriting, attention, or idea generation?"
Reading Comprehension Impact	"The learner understands better when text is read aloud." / "The learner loses meaning because reading takes too much effort."	"When someone else reads the passage, does comprehension improve?"
Reading Avoidance & Emotional Impact	"The learner avoids reading aloud." / "The learner becomes upset or frustrated during reading tasks."	"Is the avoidance due to difficulty, embarrassment, attention, anxiety, or lack of interest?"
Developmental & Family History	"Reading and spelling concerns have been present for more than 6 months." / "There is family history of dyslexia or literacy difficulties."	"Were difficulties present despite regular instruction?" / "Has the learner received previous intervention?"

Dyslexia Decision Logic

Evidence Pattern	Interpretation
Multiple Core Indicators in decoding, word reading, fluency, and spelling	Strong dyslexia hypothesis
Literacy difficulty + attention concerns	Run differential questions to separate dyslexia from ADHD-related reading issues
Reading weak only in English with strong first-language literacy	Consider language exposure factor
Reading avoidance without decoding/spelling indicators	Do not treat as strong dyslexia evidence
Strong oral comprehension but weak independent reading	Supports dyslexia hypothesis
Weak reading + weak maths + weak attention	Human review / broader assessment consideration

Recommended Rule.

For high dyslexia risk, require the following:

Requirement	Rule
Core indicators	At least 2 Core dyslexia indicators
Supporting evidence	At least 1 Supporting indicator
Consistency	Evidence appears across more than one literacy area
Differential check	ADHD/language/anxiety explanations are not stronger
Confidence	Medium or High

ADHD Domain Investigation Framework

Unlike Dyslexia, where the hallmark features revolve around literacy, ADHD is characterised by **persistent patterns of inattention and/or hyperactivity-impulsivity** that interfere with functioning across settings.

The investigation therefore seeks to determine whether observed behaviours are consistent, pervasive, developmentally inappropriate, and better explained by ADHD than by alternative factors such as anxiety, learning difficulties, fatigue, or environmental influences.

ADHD Investigation Framework

Investigation Area	BEC	Clinical Purpose	Evidence Strength
1. Sustained Attention Regulation	BEC06	Determine whether the learner can maintain attention during age-appropriate tasks.	Core
2. Behavioural Inhibition (Impulsivity)	BEC07	Determine whether the learner can regulate impulsive responses.	Core
3. Hyperactivity & Motor Regulation	BEC08	Determine whether excessive movement or restlessness is present.	Core
4. Executive Functioning & Organisation	BEC09	Determine whether the learner can plan, organise and complete tasks.	Core
5. Working Memory	BEC10	Determine whether information can be held and manipulated during learning activities.	Supporting
6. Task Persistence	BEC14	Determine whether the learner sustains effort until completion.	Supporting
7. Functional Impact Across Settings	BEC14	Determine whether behaviours occur in multiple environments (home, school, tuition, etc.).	Core
8. Emotional & Self-Regulation	BEC14	Determine whether emotions interfere with learning and behaviour.	Supporting
9. Developmental & Family History	BEC15	Determine onset, persistence and family history.	Contextual

Investigation 1: Sustained Attention Regulation

Clinical Purpose

Determine whether the learner is able to maintain attention on activities that require sustained mental effort.

Importance

Attention should fluctuate appropriately. The question is not “Can they pay attention?”

The question is “Can they regulate attention when needed?”

Behavioural Questions

- The learner loses focus during lessons.
- The learner requires frequent reminders to stay on task.
- The learner appears to daydream during activities.
- The learner becomes distracted by surrounding noises or movement.

AI Differential Questions

- Does this happen during enjoyable activities as well?
- Is attention better during one-to-one support?
- Does fatigue make the behaviour worse?
- Does the learner remain attentive when using screens or preferred activities?

Differential Diagnosis

Consider

- Dyslexia (attention lost because reading is difficult)
- Anxiety
- Sleep difficulties
- Hearing problems
- Boredom
- Environmental distractions

Investigation 2: Behavioural Inhibition (Impulsivity)

Clinical Purpose

Determine whether the learner is able to pause, think, and regulate behaviour before acting.

Behavioural Questions

- Interrupts conversations.
- Calls out answers before questions are completed.
- Difficulty waiting for their turn.
- Acts before considering consequences.

AI Clarification

- Does this happen in most environments?
- Is the learner aware afterwards that they interrupted?
- Is the behaviour occasional or persistent?

Differential Diagnosis

Consider

- ASD
- Emotional dysregulation
- Immaturity
- High excitement
- Anxiety

Investigation 3: Hyperactivity & Motor Regulation

Behavioural Questions

- Constantly fidgets.
- Leaves seat unnecessarily.
- Difficulty remaining still.
- Appears constantly "on the go."

Differential Questions

- Does this happen even during favourite activities?
- Is movement purposeful or restless?
- Does exercise reduce the behaviour?

Investigation 4: Executive Functioning

Behavioural Questions

- Frequently loses belongings.
- Starts work but forgets to finish.
- Cannot organise school materials.
- Forgets homework.
- Difficulty planning assignments.

AI Clarification

- Does organisation improve with reminders?
- Does the learner know what to do but struggle to begin?
- Does the learner become overwhelmed by multi-step tasks?

Differential Diagnosis

Could be

- ADHD
- Dyslexia
- Working memory weakness
- Anxiety

Investigation 5: Working Memory

Working Memory

Behavioural Questions

- Difficulty remembering several instructions.
- Needs instructions repeated.
- Loses track during long explanations.
- Forgets what they were doing midway through a task.

Notice...

Working memory overlaps with Dyslexia.

Different evidence strengths.

Investigation 6: Task Persistence

This is where we separate ADHD from motivation.

Questions

- Gives up easily.
- Starts enthusiastically but does not finish.
- Needs frequent encouragement.
- Leaves work incomplete.

AI asks

Does the learner stop because they are distracted or because the work is difficult?

Investigation 7: Functional Impact Across Settings

DSM-5-TR requires symptoms to occur across **multiple settings**.

Questions

- Behaviours occur both at school and at home.
- Multiple adults have noticed similar concerns.
- Difficulties affect friendships or daily routines.

AI Clarification

Does the behaviour only occur during homework?

If yes, then maybe not ADHD.

Investigation 8: Emotional Regulation

Questions

- Easily frustrated.
- Emotional reactions seem much bigger than expected.
- Difficulty calming down after disappointment.
- Mood changes interfere with learning.

Differential Could indicate • ADHD • Anxiety • Trauma • Mood disorder
Investigation 9: Developmental History
Questions • Difficulties noticed before age 12. • Family history of ADHD. • Concerns present for several years. • Previous intervention.

ADHD Decision Logic

Evidence Pattern	Interpretation
Strong evidence across attention regulation, executive functioning, and behavioural inhibition in multiple settings	Strong ADHD hypothesis
Poor attention only during reading or spelling	Consider dyslexia before ADHD
Attention difficulties only during mathematics	Consider dyscalculia before ADHD
Behaviour occurs only at school or only at home	Gather more evidence before recommending ADHD assessment
Strong executive functioning difficulties with minimal hyperactivity	Possible predominantly inattentive presentation; continue investigation
Hyperactivity present without significant inattention or impairment	Consider developmental stage or alternative explanations; human review if needed

High ADHD Risk Rule

Requirement	Rule
Core indicators	At least 2 Core investigations show consistent evidence
Functional impact	Evidence present across more than one setting (e.g., home and school)
Supporting evidence	At least 1 Supporting investigation strengthens the hypothesis
Differential check	Dyslexia, anxiety, sleep, language, and environmental factors do not better explain the behaviours
Confidence	Medium or High

Cross-Setting Consistency Gate

Before the engine can recommend an ADHD assessment, it should ask:

"Have similar behaviours been observed by more than one adult or in more than one setting?"

This becomes a mandatory decision gate because ADHD, by definition, involves persistent patterns across contexts. If the behaviours are reported only during homework or only in one classroom, the Decision Engine should lower its confidence and investigate alternative explanations before recommending referral.

Dyscalculia Domain Investigation Framework

Developmental Dyscalculia is characterised by persistent difficulties in understanding numbers, learning arithmetic facts, performing calculations, and mathematical reasoning, despite appropriate instruction.

The investigation therefore seeks evidence of fundamental difficulties in number sense, numerical processing, and mathematical cognition, rather than simply poor mathematics performance.

Dyscalculia Investigation Framework

Investigation Area	BEC	Clinical Purpose	Evidence Strength
1. Number Sense & Quantity Understanding	BEC11	Determine whether the learner understands what numbers represent.	Core
2. Number Recognition & Symbol Processing	BEC11	Determine whether the learner accurately recognises and interprets numerical symbols.	Core
3. Arithmetic Fact Retrieval	BEC12	Determine whether basic number facts become automatic with learning.	Core
4. Calculation Procedures	BEC12	Determine whether the learner can perform calculation steps accurately and consistently.	Supporting
5. Mathematical Reasoning & Problem Solving	BEC13	Determine whether mathematical concepts can be applied to real-world situations.	Supporting
6. Working Memory During Mathematics	BEC10	Determine whether temporary memory limitations interfere with mathematical tasks.	Supporting
7. Mathematical Avoidance & Emotional Impact	BEC14	Determine whether anxiety or avoidance results from persistent mathematical difficulty.	Contextual / Supporting
8. Functional Mathematical Skills	BEC13	Determine whether mathematics difficulties affect everyday activities such as time, money and estimation.	Core
9. Developmental & Family History	BEC15	Determine persistence, onset and family history.	Contextual

Investigation 1: Number Sense & Quantity Understanding

Clinical Purpose

Determine whether the learner understands quantities and relationships between numbers.

Behavioural Questions

- Has difficulty deciding which of two numbers is larger.
- Has difficulty estimating quantities.
- Frequently counts objects one by one instead of recognising groups.
- Appears unsure about what numbers represent.

AI Differential Questions

- Is the learner able to compare numbers without counting?
- Does this occur even with small numbers?
- Has the learner had sufficient opportunities to learn these concepts?

Differential Diagnosis

Consider:

- Limited mathematics instruction
- Intellectual disability
- English language proficiency (wording of problems)
- Anxiety

Investigation 2: Number Recognition & Symbol Processing

Behavioural Questions

- Confuses numerical symbols.
- Reverses numbers during reading or writing.
- Reads numbers incorrectly.
- Frequently mixes up similar-looking numbers.

AI Clarification

- Are mistakes consistent or occasional?
- Does the learner recognise numbers better when spoken aloud?
- Are errors limited to larger numbers?

Investigation 3: Arithmetic Fact Retrieval

This is the dyscalculia equivalent of decoding in dyslexia.

Behavioural Questions

- Still relies heavily on fingers for simple calculations.
- Has difficulty recalling basic addition facts.
- Needs to work out the same calculation repeatedly.
- Arithmetic facts never seem to become automatic.

AI Differential Questions

- Can the learner explain the strategy used?
- Does accuracy improve with extra time?
- Does forgetting happen across all operations?

Investigation 4: Calculation Procedures

Behavioural Questions

- Frequently forgets calculation steps.
- Loses place during long calculations.
- Makes procedural errors.
- Difficulty carrying numbers correctly.

Differential Diagnosis

Could also indicate

- ADHD
- Working memory weakness
- Inadequate instruction

Investigation 5: Mathematical Reasoning

Behavioural Questions

- Has difficulty understanding word problems.
- Cannot decide which operation to use.
- Difficulty applying mathematics to unfamiliar situations.
- Understands individual numbers but struggles with mathematical concepts.

AI Clarification

- Does the learner understand the language of the question?
- Can the learner solve the same problem using concrete objects?

Investigation 6: Working Memory During Mathematics

Behavioural Questions

- Forgets numbers halfway through calculations.
- Loses track during multi-step mathematical procedures.
- Needs repeated reminders of previous calculation steps.

Notice that this overlaps with ADHD. Different evidence strengths.

Investigation 7: Mathematics Avoidance & Emotional Impact

Questions

- Avoids mathematics whenever possible.
- Becomes anxious during maths lessons.
- Gives up quickly during calculations.
- Appears to lack confidence with numbers.

AI asks

Is the learner avoiding mathematics because they cannot understand the numbers, or because they fear making mistakes?

Investigation 8: Functional Mathematics

Questions

- Difficulty telling time.
- Difficulty estimating change.
- Difficulty comparing prices.
- Difficulty understanding measurements.
- Difficulty judging distances or quantities.

Investigation 9: Developmental History

Questions

- Difficulties have been present for several years.
- Mathematics has consistently been the greatest concern.
- Family history of mathematics difficulties.
- Previous intervention.

Dyscalculia Decision Logic

Evidence Pattern	Interpretation
Strong evidence in number sense, arithmetic fact retrieval, and functional mathematics	Strong dyscalculia hypothesis
Difficulty only with complex school mathematics but adequate number sense	Consider gaps in instruction or curriculum rather than dyscalculia
Mathematics difficulties occur together with significant reading difficulties	Investigate whether language demands or dyslexia are contributing to performance
Errors mainly involve losing place or forgetting steps	Consider ADHD or working memory difficulties before dyscalculia
Mathematics anxiety without underlying number-processing difficulties	Investigate anxiety or confidence rather than dyscalculia
Persistent number-processing difficulties across school and daily life	Supports recommendation for dyscalculia assessment

High Dyscalculia Risk Rule

Requirement	Rule
Core indicators	At least 2 Core investigations demonstrate persistent number-processing difficulties
Functional impact	Difficulties extend beyond classroom mathematics into everyday numerical tasks
Supporting evidence	At least 1 Supporting investigation strengthens the hypothesis
Differential check	Reading difficulties, ADHD, language comprehension, instructional gaps, and anxiety do not better explain the behaviours
Confidence	Medium or High

Core Screening Question Bank

Purpose

The Core Screening Question Bank serves as the initial evidence collection stage of the DIAL Adaptive Screening Engine.

Its objective is **not to identify a specific learning difference**, but to gather broad behavioural evidence across literacy, attention, executive functioning, numeracy, and learning behaviours. This evidence is used by the Decision Engine to generate initial hypotheses and determine which domain-specific investigation should be conducted next.

Response Scale

Every behavioural question uses the same response scale.

Response	Score
Never	0
Rarely	1
Sometimes	2
Often	3
Almost Always	4

Core Screening Questions

ID	Behavioural Evidence Category	Parent Version	Teacher Version	Student Version	Primary Investigation
CQ01	Reading	My child finds reading more difficult than children of the same age.	The learner finds reading more difficult than classmates.	I find reading harder than most of my classmates.	Literacy
CQ02	Reading Behaviour	My child avoids reading whenever possible.	The learner avoids reading activities.	I try to avoid reading whenever I can.	Literacy
CQ03	Writing	My child has difficulty spelling familiar words correctly.	The learner frequently spells familiar words incorrectly.	I find it difficult to spell words I already know.	Literacy
CQ04	Mathematics	My child finds mathematics more difficult than children of the same age.	The learner struggles with mathematics compared with classmates.	I find mathematics harder than most of my classmates.	Numeracy
CQ05	Attention	My child has difficulty staying focused on schoolwork.	The learner frequently loses focus during lessons.	I find it hard to stay focused in class.	ADHD

CQ06	Executive Function	My child frequently loses school materials or forgets what to bring.	The learner frequently loses belongings or forgets materials.	I often lose or forget my school things.	ADHD
CQ07	Working Memory	My child has difficulty remembering several instructions given one after another.	The learner forgets multi-step instructions.	I forget instructions when there are many steps.	ADHD / Dyslexia
CQ08	Learning Behaviour	Homework often becomes frustrating for my child.	The learner becomes frustrated during learning activities.	I get frustrated when doing schoolwork.	General
CQ09	Processing Speed	My child needs much more time than others to complete schoolwork.	The learner completes work much more slowly than peers.	I need much more time than my classmates to finish my work.	General
CQ10	Reading Confidence	My child lacks confidence when reading aloud.	The learner appears uncomfortable reading aloud.	I don't feel confident reading aloud.	Dyslexia
CQ11	Mathematics Confidence	My child lacks confidence when working with numbers.	The learner appears anxious when working with mathematics.	I don't feel confident doing maths.	Dyscalculia

CQ12	Attention Regulation	My child is easily distracted by surrounding noises or activities.	The learner is easily distracted in class.	I get distracted easily.	ADHD
CQ13	Organisation	My child forgets homework or daily responsibilities.	The learner frequently forgets homework or responsibilities.	I often forget homework or things I need to do.	ADHD
CQ14	General Learning	My child seems to require much more support than other children.	The learner requires significantly more support than peers.	I often need more help than my classmates.	General
CQ15	Parent/Teacher Concern	Which area concerns you the most? (Reading, Writing, Mathematics, Attention, Organisation, Unsure)	Same	Which subject or activity feels hardest for you?	Routing only (not scored)

Routing Output

The Decision Engine should not produce Dyslexia score: 67

Instead, it produces something like:

Hypothesis	Evidence	Confidence
Dyslexia	Strong	Medium
ADHD	Moderate	Medium
Dyscalculia	Weak	High

Then, AI decides “Which uncertainty should I reduce next?”

Domain Question Bank: Dyslexia

Investigation 1: Phonological Processing

Clinical Purpose

Determine whether the learner demonstrates persistent difficulty recognising, manipulating and remembering speech sounds.

Behavioural Evidence Category: BEC01

Evidence Strength: Core

ID	Behavioural Question	Intent	AI Clarification Prompt
DYX01	The learner has difficulty sounding out unfamiliar words.	Assess phonological decoding.	Can you describe what happens when your child encounters a new word?
DYX02	The learner confuses words that sound similar.	Detect phonological discrimination difficulties.	Does this happen when listening, reading, or both?
DYX03	The learner has difficulty identifying the first or last sound in a word.	Assess phonological awareness.	Can your child identify rhyming words or break words into sounds?
DYX04	The learner struggles to blend individual sounds together to read a word.	Assess phoneme blending.	Does your child guess the word instead of sounding it out?
DYX05	The learner finds rhyming activities unusually difficult.	Assess early phonological awareness.	Has this difficulty been present since preschool or early primary school?

Differential Considerations

Possible alternative explanations:

- Limited phonics instruction
- English as an additional language
- Speech/language disorder
- Hearing difficulties

Investigation 2: Word Reading Accuracy

Clinical Purpose: Determine whether the learner accurately recognises printed words.

Behavioural Evidence Category: BEC02

Evidence Strength: Core

ID	Behavioural Question	Intent	AI Clarification
DYX06	The learner frequently guesses words instead of reading them accurately.	Detect guessing strategy.	Are guesses visually similar to the correct word?
DYX07	The learner substitutes one word for another while reading.	Assess word recognition accuracy.	Does the learner usually notice the mistake?
DYX08	The learner skips small words while reading.	Detect reading accuracy problems.	Does this happen mainly when reading aloud?
DYX09	The learner rereads sentences several times.	Detect inaccurate decoding affecting comprehension.	Is rereading due to misunderstanding or word recognition?
DYX10	The learner misreads familiar words that should already be known.	Assess automatic word recognition.	Does accuracy improve when reading slowly?

Investigation 3: Reading Fluency

Behavioural Evidence Category: BEC03

Evidence Strength: Core

ID	Behavioural Question	Intent	AI Clarification
DYX11	Reading aloud is noticeably slower than expected for the learner's age.	Assess fluency.	Is reading slow because words are difficult to recognise?
DYX12	The learner pauses frequently while reading.	Detect decoding interruptions.	Does the learner stop to sound out words?
DYX13	Reading sounds effortful rather than smooth.	Assess automaticity.	Does fluency improve with familiar texts?
DYX14	The learner loses their place while reading.	Detect tracking difficulties.	Is this due to attention or word recognition?
DYX15	Reading becomes tiring very quickly.	Assess cognitive load during reading.	After how many minutes does this usually happen?

Investigation 4: Spelling Consistency

Behavioural Evidence Category: BEC05

Evidence Strength: Core

ID	Behavioural Question	Intent
DYX16	The learner spells the same word differently in the same piece of writing.	Detect unstable spelling representations.
DYX17	Spelling improves very little despite repeated practice.	Assess persistence of spelling difficulties.
DYX18	The learner spells words based on how they sound.	Assess phonetic spelling patterns.
DYX19	The learner frequently omits or rearranges letters.	Assess orthographic processing.
DYX20	The learner struggles to remember common spelling patterns.	Assess long-term spelling knowledge.

Investigation 5: Written Expression

Behavioural Evidence Category: BEC05

Evidence Strength: Supporting

ID	Behavioural Question	Intent
DYX21	The learner has good ideas but struggles to express them in writing.	Separate language ability from written output.
DYX22	Written work is much shorter than verbal explanations.	Detect writing avoidance due to literacy demands.
DYX23	The learner spends considerable effort on spelling, leaving little attention for ideas.	Assess cognitive load during writing.
DYX24	Sentence writing is slower than expected.	Detect writing fluency difficulties.
DYX25	The learner avoids extended writing tasks.	Assess writing avoidance.

Investigation 6: Reading Comprehension

Behavioural Evidence Category: BEC04

Evidence Strength: Supporting

ID	Behavioural Question	Intent
DYX26	The learner understands stories better when someone else reads them aloud.	Distinguish decoding from language comprehension.
DYX27	The learner forgets what was read because reading required too much effort.	Assess decoding impact on comprehension.
DYX28	The learner can explain ideas verbally better than through independent reading.	Compare oral and reading comprehension.
DYX29	Reading accuracy affects understanding of the passage.	Assess literacy impact.
DYX30	Comprehension improves significantly when decoding demands are reduced.	Differentiate comprehension from decoding.

Investigation 7: Reading Avoidance & Emotional Impact

Behavioural Evidence Category: BEC14

Evidence Strength: Supporting / Contextual

ID	Behavioural Question	Intent
DYX31	The learner avoids reading aloud whenever possible.	Detect reading avoidance.
DYX32	Reading activities cause noticeable frustration.	Assess emotional impact.
DYX33	The learner lacks confidence when reading.	Assess self-perception.
DYX34	The learner becomes anxious before reading tasks.	Assess reading anxiety.
DYX35	The learner prefers others to read for them.	Assess avoidance behaviour.

Investigation 8: Developmental & Family History

Behavioural Evidence Category: BEC15

Evidence Strength: Contextual

ID	Behavioural Question	Intent
DYX36	Reading difficulties have been present for more than six months despite appropriate instruction.	Assess persistence.
DYX37	Similar literacy difficulties exist within the family.	Family history.
DYX38	Reading difficulties were noticed from the early years of schooling.	Developmental onset.
DYX39	Previous interventions produced only limited improvement.	Response to intervention.
DYX40	Reading remains significantly below expectations despite ongoing support.	Longitudinal evidence.

Investigation Priority

Priority	Investigation	Why First?
1	Phonological Processing	Hallmark feature
2	Word Reading Accuracy	Confirms decoding impact
3	Reading Fluency	Measures functional impact
4	Spelling	Confirms orthographic difficulties
5	Reading Comprehension	Determines downstream effects
6	Written Expression	Measures academic impact
7	Reading Avoidance	Emotional consequences
8	Developmental History	Confirms persistence

Investigation Entry / Exit Criteria		
Entry Criteria	Exit Criteria	Escalation Criteria
When should this investigation begin? Example Reading Fluency Start if - Reading Difficulty = High OR - Reading Avoidance = High	When do we stop? Example Stop Reading Fluency Investigation when - Two Core Indicators collected OR - Confidence > 90% OR - Investigation contradicted	Example Reading Fluency ↓ Evidence conflicts with Word Reading ↓ Escalate ↓ Differential Investigation

Investigation Orchestration Framework

Purpose

The **Investigation Orchestration Framework** defines how the DIAL Adaptive Screening Engine decides **what to investigate next, which question to ask, when enough evidence has been collected**, and **when human review is required**.

It ensures the screener behaves like an adaptive reasoning system rather than a fixed questionnaire.

Orchestration Flow:

Stage	What Happens
1. Start Screening	User completes consent, learner profile, and core questions.
2. Generate Initial Hypotheses	Decision Engine ranks possible concerns: Dyslexia, ADHD, Dyscalculia.
3. Select Primary Investigation Path	Highest-evidence hypothesis becomes the first investigation path.
4. Choose Investigation Area	System selects the most clinically important area to investigate first.
5. Ask Best Question	AI selects a question from the approved repository.
6. Update Evidence	Response updates evidence strength, confidence, and risk.
7. Check Exit Criteria	If enough evidence is collected, move to next investigation or recommendation.
8. Run Differential Investigation	If evidence is mixed, AI asks questions to distinguish competing explanations.
9. Recommend Pathway	System recommends one primary assessment pathway.
10. Trigger Human Review	Complex, unclear, or high-risk cases are routed to psychologist/admin review.

Investigation Object

Each investigation should be stored as a structured object.

Field	Description	Example
Investigation ID	Unique identifier	DYX_INV_01
Investigation Name	Area being investigated	Phonological Processing
Domain	Main hypothesis	Dyslexia
BEC Mapping	Behavioural Evidence Category	BEC01
Priority Level	Order of importance	1
Evidence Strength	Core / Supporting / Contextual	Core
Entry Conditions	When investigation begins	Dyslexia evidence is moderate/high
Minimum Questions	Minimum evidence to collect	2
Maximum Questions	Prevents over-questioning	5
Exit Conditions	When to stop this investigation	2 Core indicators confirmed
Escalation Rules	When to switch or review	Conflicting ADHD evidence
Differential Considerations	Alternative explanations	ADHD, language exposure, anxiety
AI Strategy	What AI should do	Clarify decoding vs attention issue

Dyslexia Investigation Order

Priority	Investigation Area	Why This Order
1	Phonological Processing	Hallmark dyslexia evidence.
2	Word Reading Accuracy	Confirms printed-word reading difficulty.
3	Decoding Unfamiliar Words	Separates guessing from phonics-based decoding.
4	Reading Fluency	Shows functional impact during reading.
5	Spelling Consistency	Strengthens literacy-pattern evidence.
6	Reading Comprehension Impact	Checks whether comprehension is affected by decoding load.
7	Written Expression	Checks academic output impact.
8	Reading Avoidance & Emotional Impact	Contextualises frustration and avoidance.
9	Developmental & Family History	Confirms persistence and background risk.

ADHD Investigation Order

Priority	Investigation Area	Why This Order
1	Sustained Attention Regulation	Central ADHD evidence.
2	Executive Functioning & Organisation	Often highly visible in school functioning.
3	Behavioural Inhibition / Impulsivity	Distinguishes ADHD from general inattentiveness.
4	Hyperactivity & Motor Regulation	Captures hyperactive presentation.
5	Working Memory	Supports attention/executive-function profile.
6	Task Persistence	Checks effort regulation and task completion.

7	Functional Impact Across Settings	Mandatory ADHD confidence gate.
8	Emotional & Self-Regulation	Captures regulation difficulties.
9	Developmental & Family History	Checks onset, persistence, and family context.

Dyscalculia Investigation Order

Priority	Investigation Area	Why This Order
1	Number Sense & Quantity Understanding	Core dyscalculia evidence.
2	Number Recognition & Symbol Processing	Checks symbolic number processing.
3	Arithmetic Fact Retrieval	Checks automaticity of number facts.
4	Calculation Procedures	Checks procedural accuracy.
5	Functional Mathematical Skills	Shows real-world numerical impact.
6	Mathematical Reasoning & Problem Solving	Checks concept application.
7	Working Memory During Mathematics	Separates dyscalculia from memory/attention issues.
8	Mathematical Avoidance & Emotional Impact	Contextualises anxiety and avoidance.
9	Developmental & Family History	Confirms persistence and history.

Entry and Exit Rules

Rule Type	Description
Entry Rule	Begin an investigation only when core evidence suggests the domain may be relevant.
Minimum Evidence Rule	Ask at least 2 questions for any active Core investigation.
Maximum Question Rule	Stop after 5–6 questions within one investigation to prevent fatigue.
Exit Rule	Stop an investigation when evidence is strong, weak, or contradicted enough to decide.
Escalation Rule	Move to differential investigation if another hypothesis becomes similarly strong.
Human Review Rule	Escalate if confidence remains low, evidence is mixed, or risk is high-complexity.

Example Orchestration:

Step	System State	Action
1	Core questions completed	Dyslexia evidence strongest; ADHD moderate.
2	Enter Dyslexia path	Start Phonological Processing investigation.
3	Evidence unclear	Ask decoding clarification question.
4	Strong decoding evidence found	Move to Word Reading Accuracy.
5	Attention evidence still present	Run differential question: decoding difficulty vs attention loss.
6	Dyslexia remains primary	Continue Dyslexia path.
7	Enough evidence collected	Generate Dyslexia assessment recommendation.
8	ADHD noted	Mark attention concerns as Secondary KIV.

DIAL Question Repository Schema

Purpose

The Question Repository is the central library of all approved behavioural questions used by the DIAL Adaptive Screening Engine.

Rather than functioning as a static questionnaire, the repository stores each question as a structured knowledge object containing:

- its educational purpose,
- behavioural evidence mapping,
- investigation context,
- routing rules,
- clarification prompts,
- and decision metadata.

This allows the AI to select the **most informative question** for the current investigation while remaining bounded by psychologist-approved content.